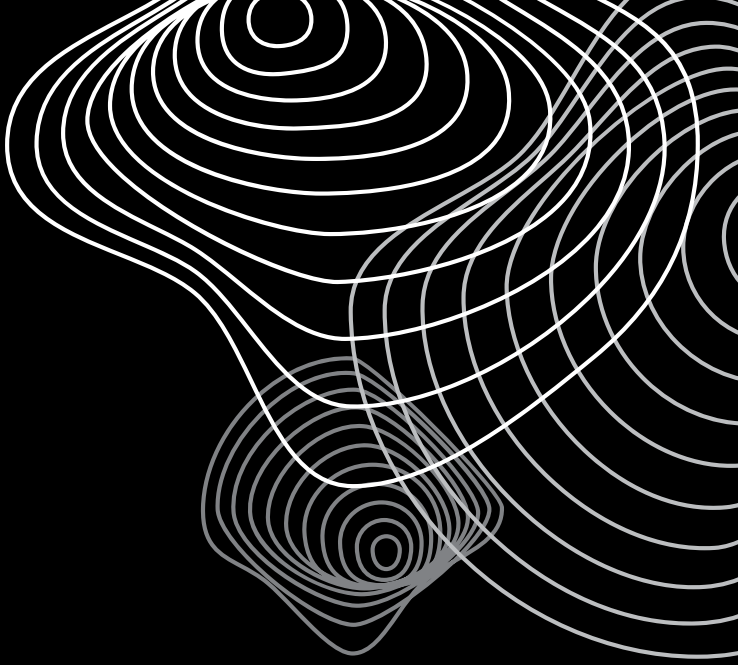




Heterogeneous Riders

TicketingHub

Modern Customer Support Ticketing System

Streamline Your Support. Delight Your Customers.

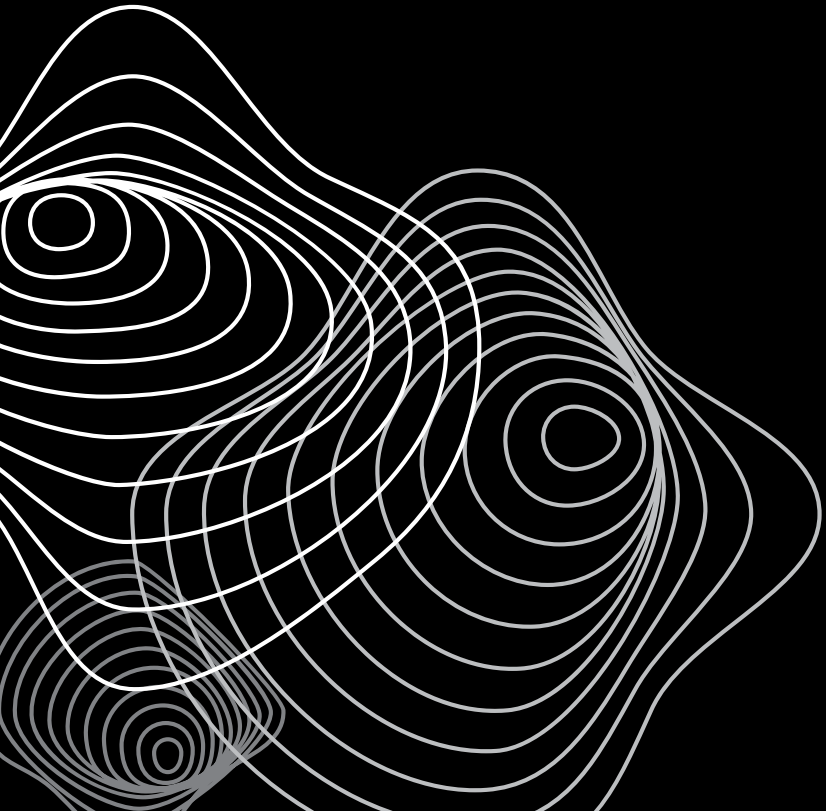
React Frontend

ASP.NET Web API

MSSQL

Secure

Scalable



TEAM



Purushothaman S



Prasanna P



Priyadharshini R



Eshwanth K

Existing Problem And Solution

The Challenge

-  **Fragmented Communication**
Multiple email threads and support channels create chaos
-  **Delayed Response Times**
Manual triage leads to slow, inconsistent customer service
-  **Disorganized Workflows**
No visibility into ticket status or agent workload
-  **Limited Reporting**
No analytics or accountability metrics

Our Solution

-  **Centralized Ticketing**
One platform for all support channels and tickets
-  **Automated Routing**
Intelligent categorization and agent assignment
-  **Real-time Notifications**
Instant notifications
-  **Actionable Analytics**
Role-based dashboards and comprehensive reporting

Expected Outcome

Customer Experience

-  **Create Tickets Easily**
Submit support tickets via web interface or email
-  **Track Status & History**
Real-time visibility into ticket progress and past interactions
-  **Real-time Notifications**
Instant messaging with support agents for quick resolution
-  **Attachments & Updates**
Attach files and receive automatic notifications

Agent/Admin Experience

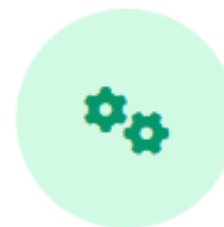
-  **Unified Dashboards**
Role-based views for agents, managers, and administrators
-  **Automated Routing**
Intelligent ticket categorization and agent assignment
-  **Smart Categorization**
Ticket classification and priority setting
-  **Reports & Analytics**
Comprehensive reporting with exports and notifications

Built on Modern Technology

Scalable architecture with enterprise-grade reliability



React Frontend
Modern responsive UI



ASP.NET Web API
Robust backend services



MSSQL Database
Secure data storage

Tech Stack

Frontend Technologies

-  **React 19**
Modern, fast, and interactive user interface library
-  **React Router**
Client-side routing for seamless navigation
-  **Redux Toolkit**
State management for predictable data flow
-  **Vite**
Ultra-fast build tool for development
-  **TypeScript**
Type-safe JavaScript for better code quality

Backend Technologies

-  **ASP.NET Web API (.NET 9)**
Robust, scalable REST API framework
-  **MSSQL Database**
Enterprise-grade relational database with EF Core
-  **MediatR (CQRS)**
Clean architecture with command query separation
-  **Email Service**
Real-time ticket creation while user send mail
-  **Swagger/OpenAPI**
Interactive API documentation and testing

How It Works – Ticket Lifecycle

From creation to resolution in 4 simple steps



- Email Integration
- AI Categorization
- Notifications
- Analytics

Folders, Component and Structure

</> Frontend Modules

-  **Authentication**
Login, register, forgot-password pages
-  **Tickets**
Ticket creation, listing, detail pages, comments
-  **Dashboard**
Dashboard UI, charts, statistics cards
-  **Configuration**
SLA policies, escalation, timers
-  **User Management**
User management, profile pages, CRUD operations
-  **Notifications**
Notification list, toast handling, settings
-  **Settings**
App-level settings/preferences UI
-  **Shared Components**
Reusable UI components, global services

Backend Components

-  **Controllers**
API endpoint controllers (thin controllers)
-  **Domain Models**
Entities, enums, and domain models
-  **Features**
Handlers, DTOs, Services (MediatR CQRS)
-  **Infrastructure**
Repositories, Email services, integrations
-  **Security**
Authentication, Authorization, JWT, Claims setup
-  **Migrations**
EF Core database migrations
-  **Extensions**
Application/service extension methods

Security & Compliance

Enterprise-grade security for your customer data



JWT Authentication

Token-based authentication with secure token management



Role-Based Authorization

Flexible permissions with admin, agent, and customer roles



Input Validation

Comprehensive validation of all user inputs



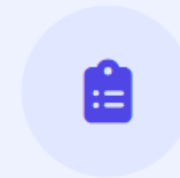
Secure File Handling

Safe attachment uploads with file type restrictions



Encrypted Data Transmission

HTTPS encryption for all data transmission



Audit Logging

Comprehensive activity tracking and audit trails for compliance

Business benefit and Future Roadmap

Immediate Benefits

-  **Improved Customer Satisfaction**
Real-time updates and faster response times
-  **50% Faster Response Times**
Automated routing and intelligent prioritization
-  **Reduced Operational Costs**
Automated workflows and reduced manual effort
-  **Real-time Insights**
Actionable analytics and performance metrics

Future Roadmap

-  **PWA Mobile Support**
Progressive Web App for mobile devices
-  **Single Sign On**
Single sign on with microsoft
-  **AI-powered ticket processing**
Natural language processing ticket processing based on history
-  **Advanced SLA Tracking**
Enhanced service level agreement monitoring
-  **Signal R**
Live notifications and chats

THANK YOU